

(10)

Algorithm

An algorithm is a step-by-step plan to solve the problem.

1. Start

2. Initialize Sample Logs: Create a list of strings where each string represents a log entry containing a Timestamp, Severity Level, Username, and Message.

3. Display Menu: Show options to the user:

- View All Logs
- Count Severity Levels (Error, Info, Warning)
- Analyze User Activity
- Filter Logs by Severity
- Exit

4. Get User Choice: Read the user's input.

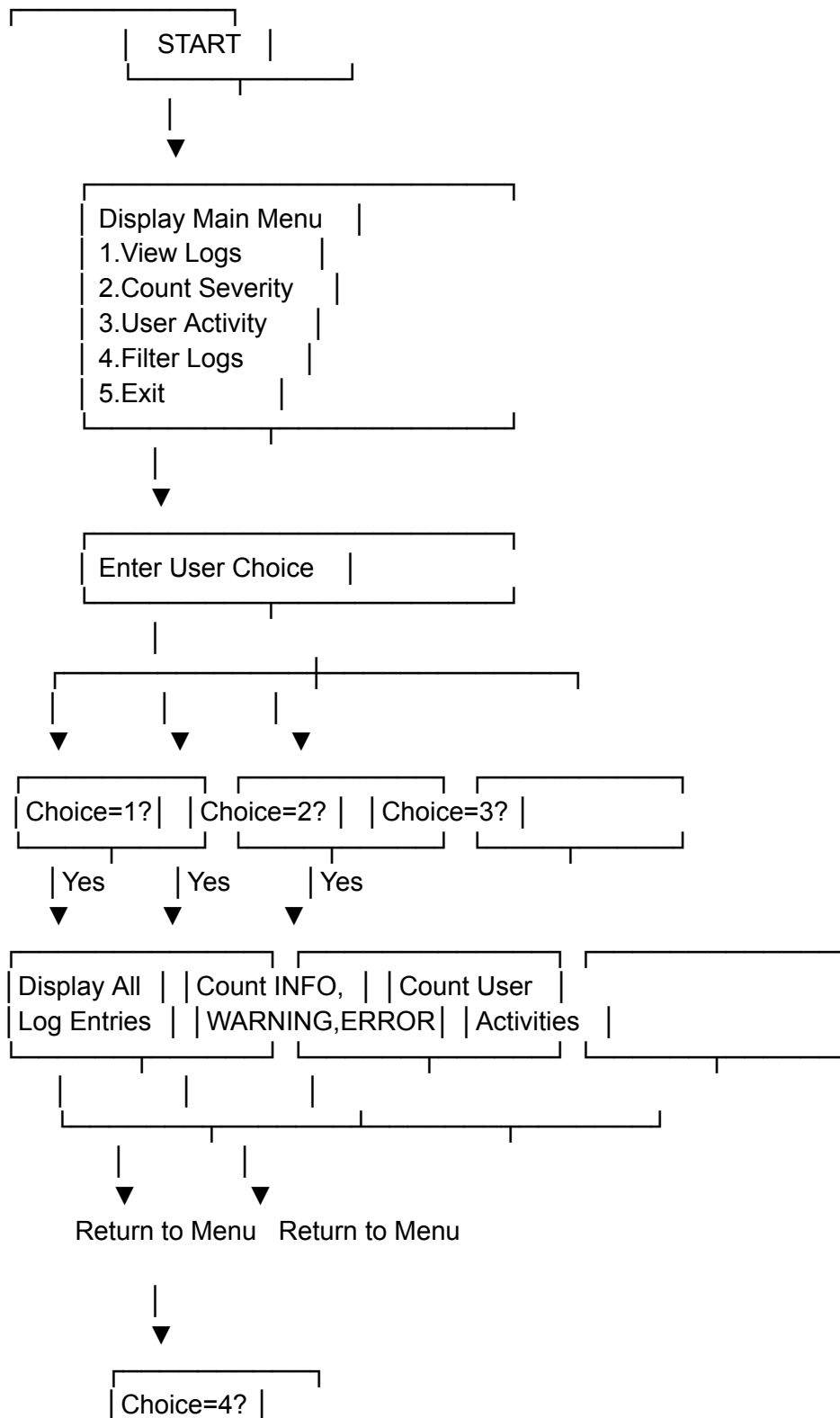
5. Process Choice:

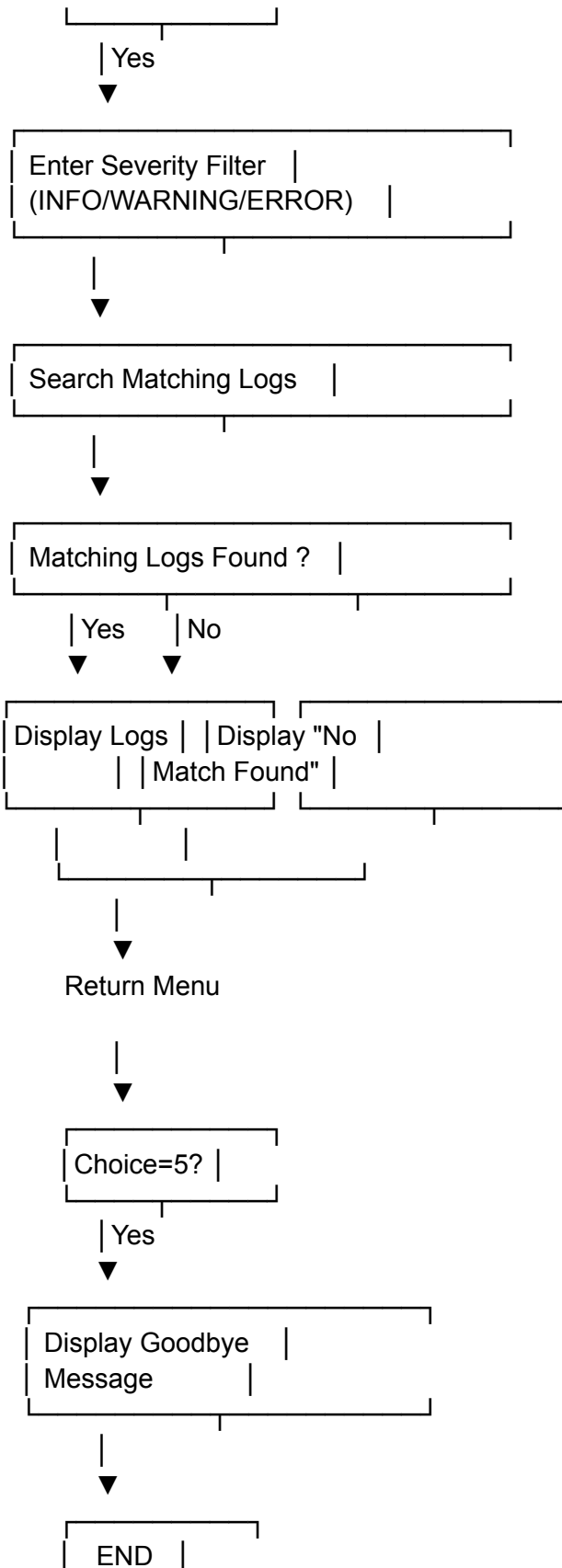
- Choice 1: Loop through the log list and print each log.
- Choice 2: Initialize an empty dictionary. Loop through each log, split the string by spaces to find the severity keyword, and count them using the dictionary.
- Choice 3: Initialize an empty dictionary. Loop through each log, extract the username, and count how many actions each user performed.
- Choice 4: Ask the user for a severity level (e.g., ERROR). Loop through logs and print only the ones that match.
- Choice 5: Print a goodbye message and terminate the loop.

6. Repeat: Go back to Step 3 unless Choice 5 is selected.

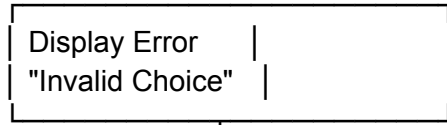
7. Stop

Flowchart





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If Invalid Choice:



Return to Menu